



Hotel Cancellation Forecast

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The Hotel Industry



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- Hotels have daily limited perishable inventory
 - Unsold rooms = Lost Revenue
 - Overselling rooms:
 - Pay for a comparable hotel
 - Transportation costs
 - Damage to reputation and loyalty



How Machine Learning Can Help

- Metric of Success: Reaching a "perfect sell"
- Predicting cancellations to optimize pricing strategies
- ML Models to solve classification problems:
 - Logistic Regression
 - Random Forest

Goal: Reliability and Generalization

The Dataset

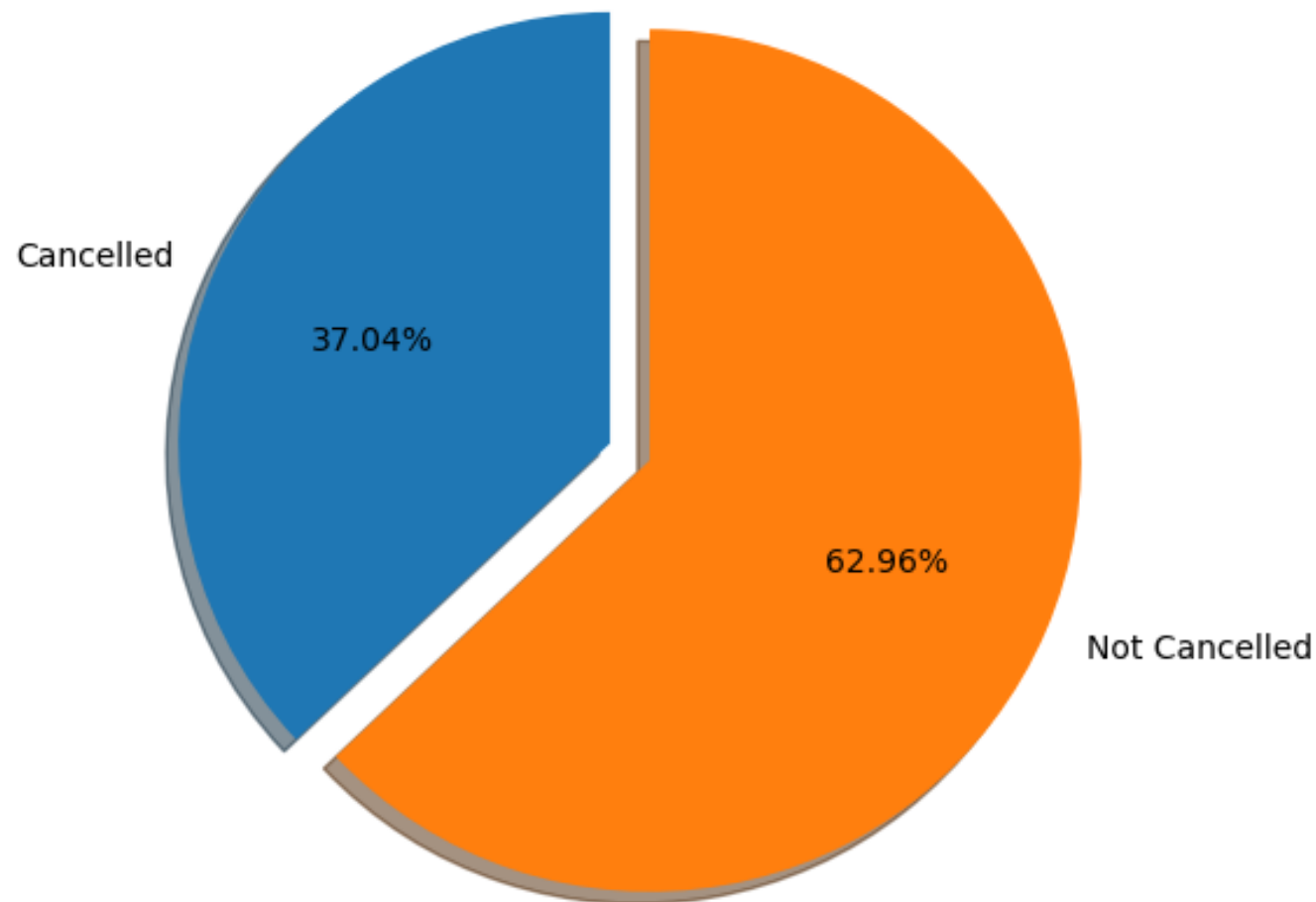
Important Columns:

- *is_canceled*
- *Market_segment*
- *Lead_time*
- *Customer_type*

Target Class Imbalance Identified:

- *Total reservations: 119,000*
- *Reservations cancelled: 44,000*

Percentage of Cancelled Reservations





Obstacles

- Overfitting – Initial models had 100% train accuracy or 100% precision
- Through exploration, corrected the target class imbalance
- Leak identified: Reservation_Status

Model Comparisons

Random Forest:

<u>Metric</u>	<u>Train</u>	<u>Test</u>	<u>Overfit Amt</u>
Accuracy	81.91%	88.80%	1.11%
AUC	90.90%	87.70%	3.20%
F1- Score	80.44%	72.26%	8.18%

XGBoost:

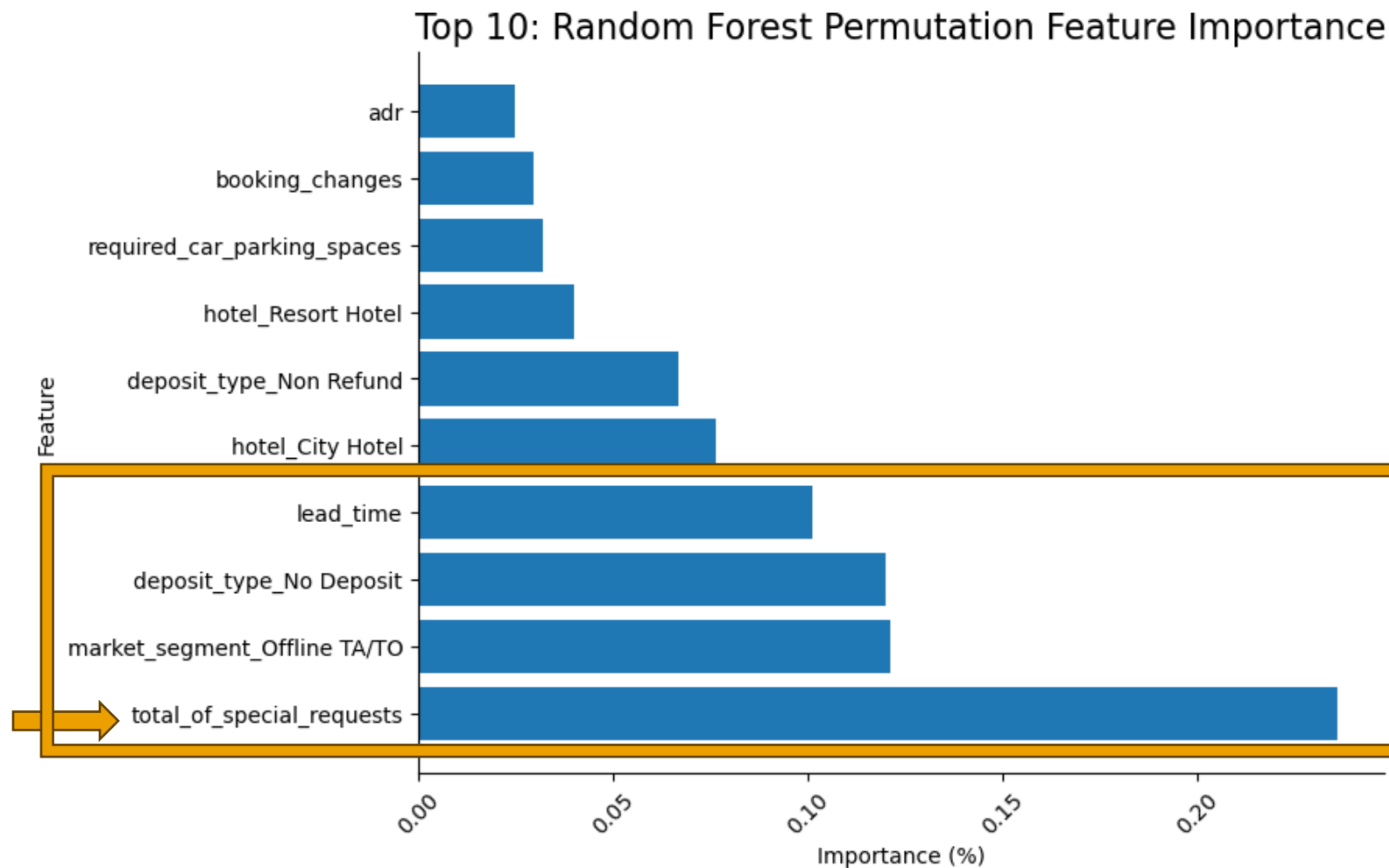
<u>Metric</u>	<u>Train</u>	<u>Test</u>	<u>Overfit Amt</u>
Accuracy	90.80	84.33%	6.47%
AUC	94.40%	90.10%	4.30%
F1- Score	90.58%	78.18%	12.40%

- XGBoost:
 - 4% higher accuracy
 - 3% higher AUC
 - 6% higher F-1 score
- XGBoost is more overfitted
- Goal: Reliability and Generalization

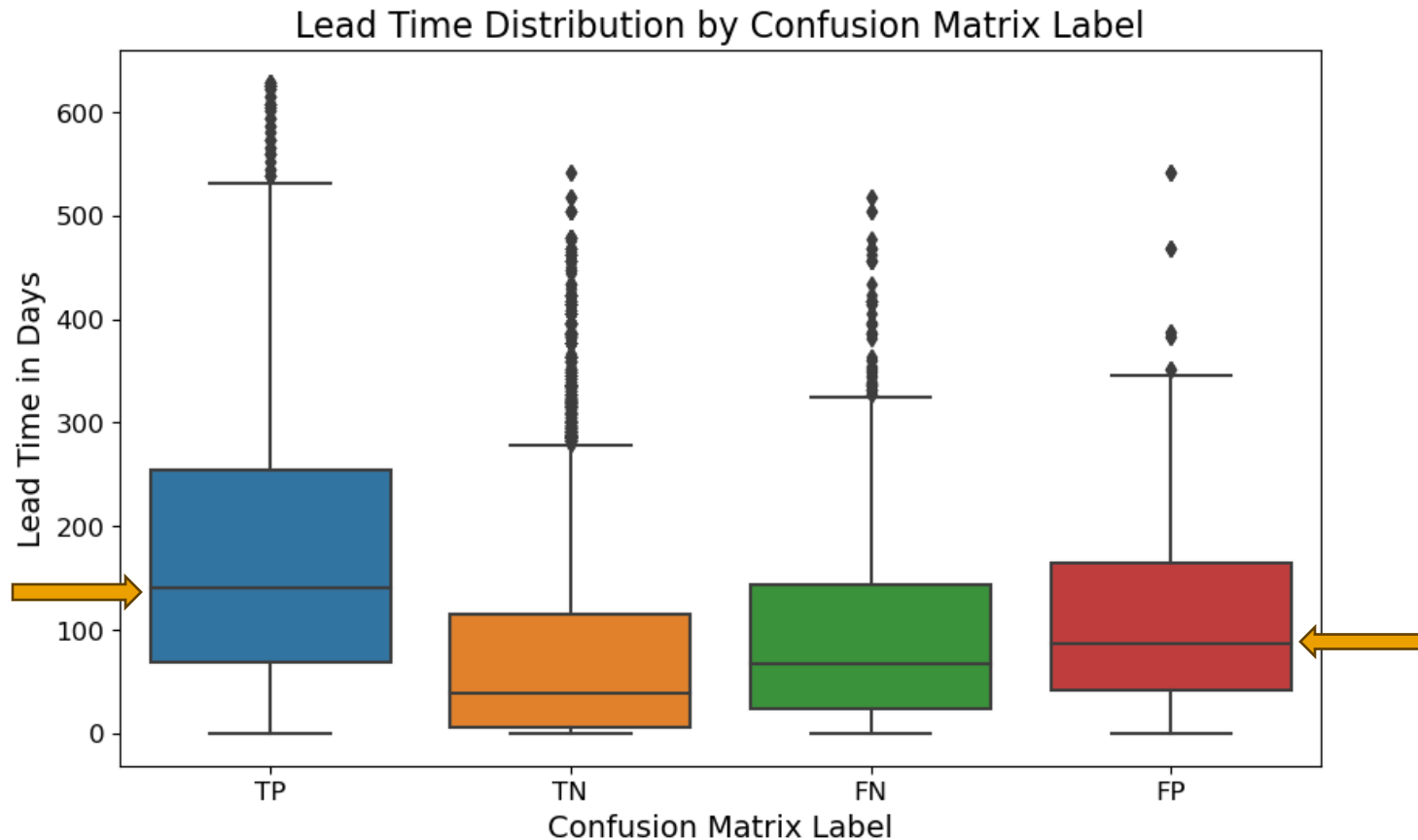
Feature Importance

Permutation Feature Importance

- Total Special Requests: 20%
- Market Segment TA/TO: 12%
- Deposit Type: No Deposit: 12%
- Lead Time: 10%



Error Analysis: Lead Time Distribution



True Positives vs. False Positive

- TP: Mean is above 100 days
- FP: Mean is below 100 days
- Model struggles with reservations that have shorter lead times (less than 100 days)

Next Steps

- Explore additional Feature Engineering to address errors
 - Binning features like Lead Time
 - Removing unnecessary features
- Predict daily occupancy
 - Explore Time Series Forecasting for Demand



Thank You

